

# Fitting A Linear Model

Code ▾

## Setup

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```
library(tidyverse)
library(patchwork)
library(ggplot2)

# Change this to your week 4 folder
setwd(r"(C:/Users/Valentin/Documents/GIT_REPS/TUHH/Applied-Statistics/week04)")
```

Note that when saving images, do it with this ratio to fit perfectly into the TUHH template:

width = 8.74, height = 5.11

## Import Data

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```
video_game_data <- read.csv("Video_Games_Sales_as_at_22_Dec_2016.csv")

clean_data <- video_game_data |>
  select(User_Score, Critic_Score, Year_of_Release, Global_Sales) |>
  mutate(User_Score = as.numeric(User_Score)*10) |>
  mutate(Year_of_Release = as.numeric(Year_of_Release)) |>
  select(where(is.numeric)) |>
  drop_na()

# Row index of the game with max sales
max_index <- which.max(clean_data$Global_Sales)

# View the entire row
clean_data[max_index, ]
```

|       | User_Score<br><dbl> | Critic_Score<br><int> | Year_of_Release<br><dbl> | Global_Sales<br><dbl> |
|-------|---------------------|-----------------------|--------------------------|-----------------------|
| 1     | 80                  | 76                    | 2006                     | 82.53                 |
| 1 row |                     |                       |                          |                       |

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```
print(length(clean_data$User_Score))
```

```
[1] 6894
```

# Data Visualization

filtering N/A:

$$N = 16719$$
$$N_{\text{filtered}} = 6894$$

## User and Critic Score

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```
long <- clean_data %>%
  select(Year_of_Release, Global_Sales, User_Score, Critic_Score) %>%
  pivot_longer(cols = c(User_Score, Critic_Score),
               names_to = "Type",
               values_to = "Score")

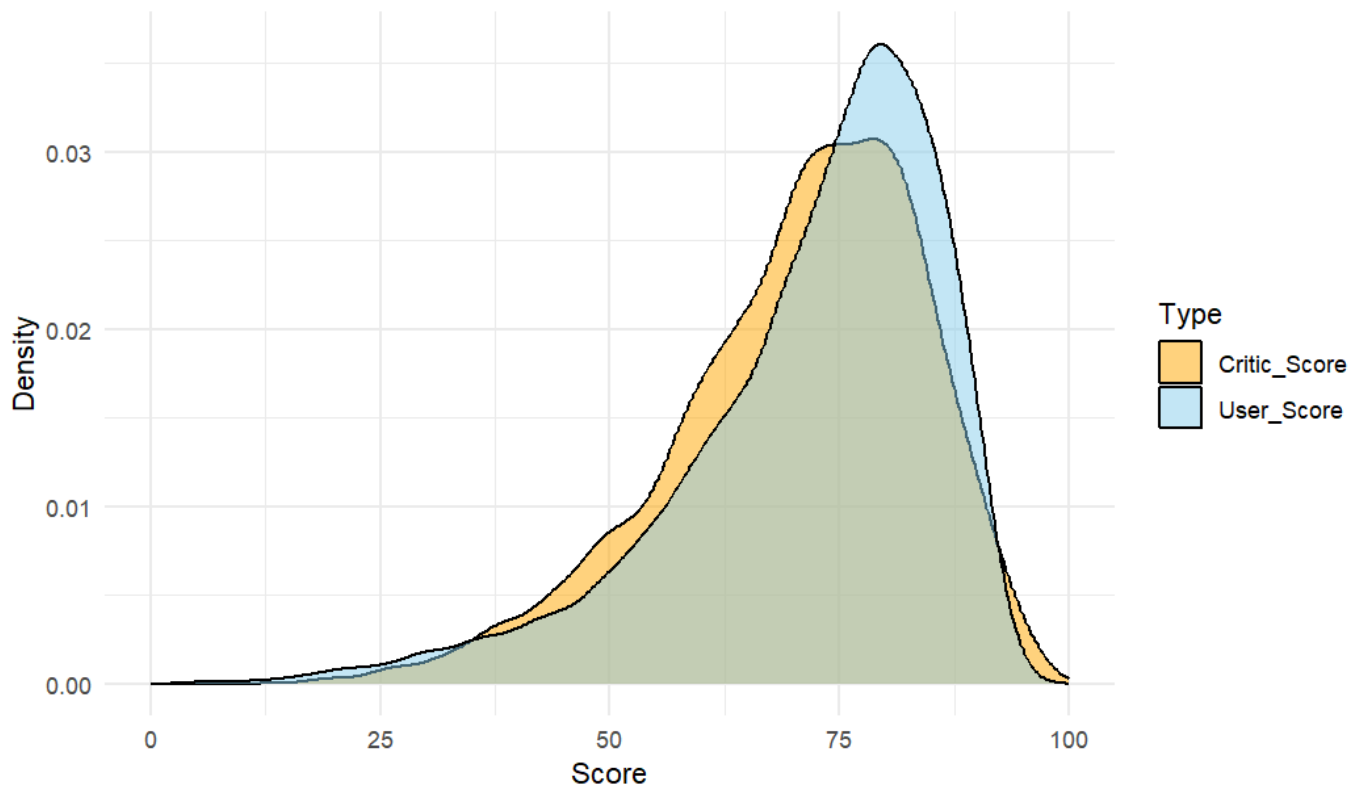
distribution_plot <- ggplot(long, aes(x = Score, fill = Type)) +
  geom_density(alpha = 0.5) +
  scale_x_continuous(limits = c(0, 100)) +
  labs(
    x = "Score",
    y = "Density",
    title = "Distribution of User vs Critic Scores"
  ) +
  scale_fill_manual(values = c("User_Score" = "skyblue", "Critic_Score" = "orange")) +
  theme_minimal()

ggsave("plots/distribution_plot.png", distribution_plot, width = 8.74, height = 5.11, dpi = 300)

distribution_plot

par(mfrow = c(1, 2))
```

## Distribution of User vs Critic Scores



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```
qqnorm(clean_data$User_Score, main = "QQ Plot: User Score", ylim = c(0, 100))
qqline(clean_data$User_Score, col = "red")
```

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```
qqnorm(clean_data$Critic_Score, main = "QQ Plot: Critic Score", ylim = c(0, 100))
qqline(clean_data$Critic_Score, col = "red")
```

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```
width_px <- 8.74 * 300
height_px <- 5.11 * 300

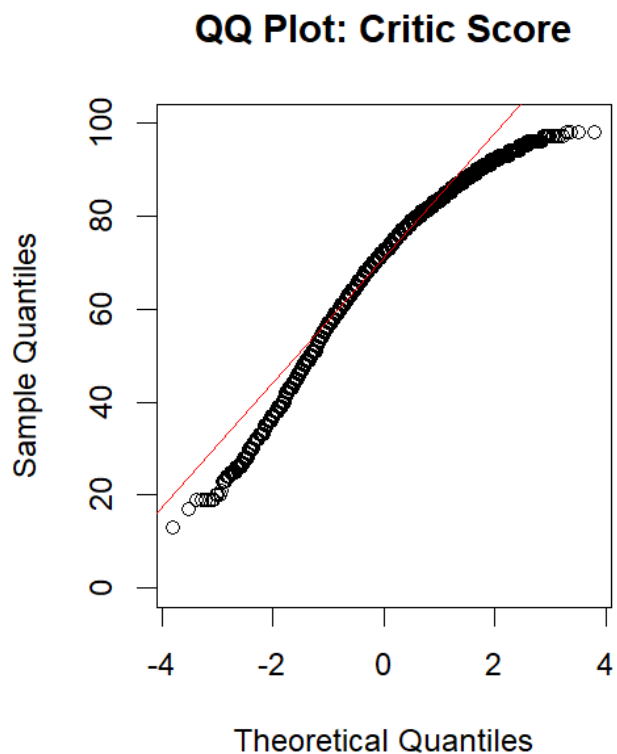
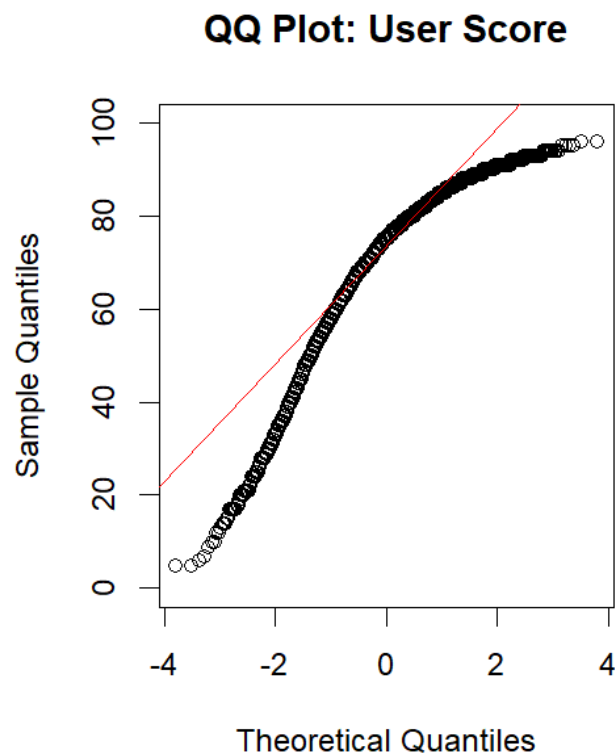
dev.copy(png, "plots/QQplots_User_Critic.png", width = width_px, height = height_px, res = 300)
```

```
png
3
```

Hide

```
dev.off()
```

```
png
2
```



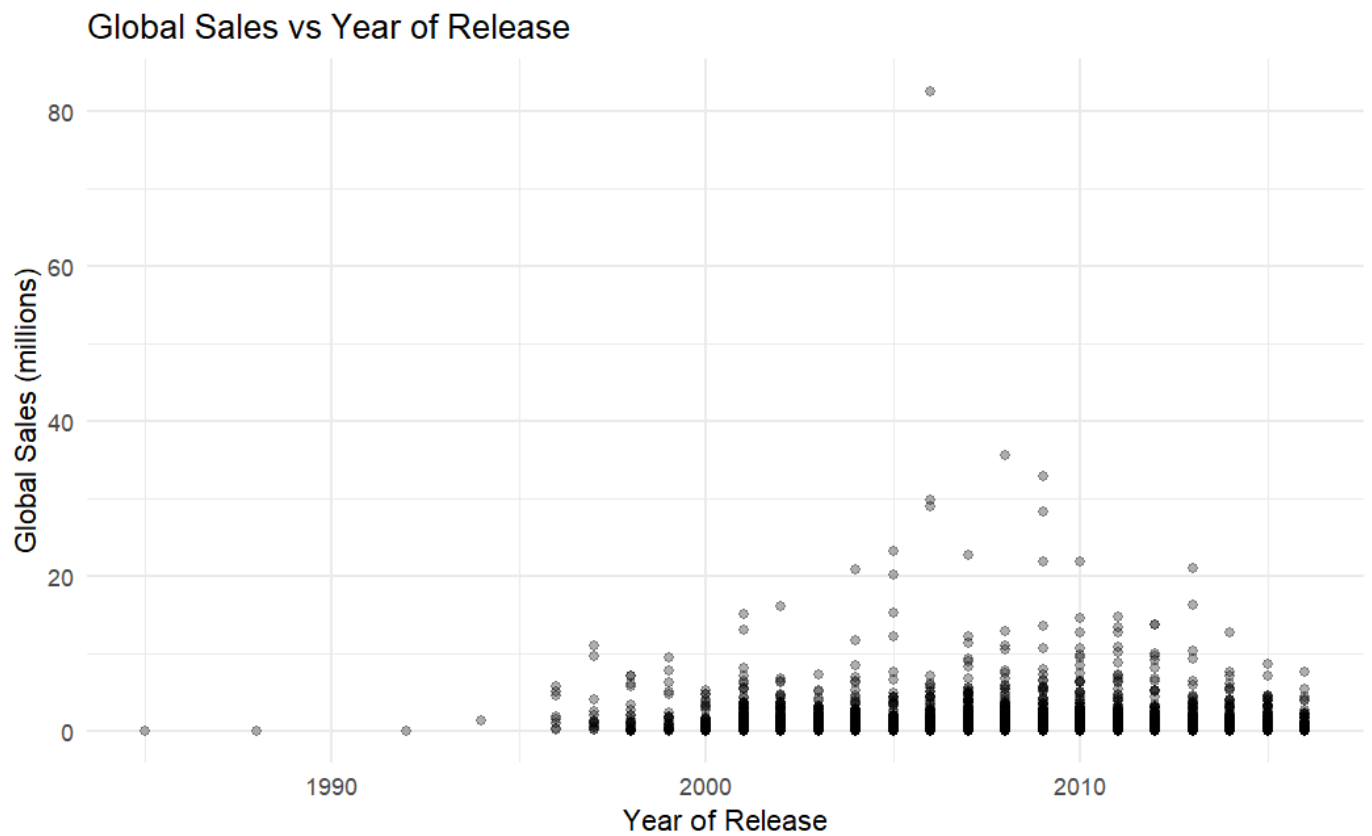
## Year of Release and Global Sales

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```
sales_release_plot <- ggplot(clean_data, aes(x = Year_of_Release, y = Global_Sales)) +
  geom_point(alpha = 0.3) +
  labs(x = "Year of Release",
       y = "Global Sales (millions)",
       title = "Global Sales vs Year of Release") +
  theme_minimal()

ggsave("plots/sales_release_plot.png", sales_release_plot, width = 8.74, height = 5.11, dpi =
300)

sales_release_plot
```



Incomplete List of Games, Most Games are not on there, such as the Pc version of Minecraft missing and the list only including games up to 2016 The 2006 Outlier is Wii U Sports

## Rating per Year and Sales

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```

# Compute average per year
avg_year <- long %>%
  group_by(Year_of_Release, Type) %>%
  summarise(mean_score = mean(Score, na.rm = TRUE))

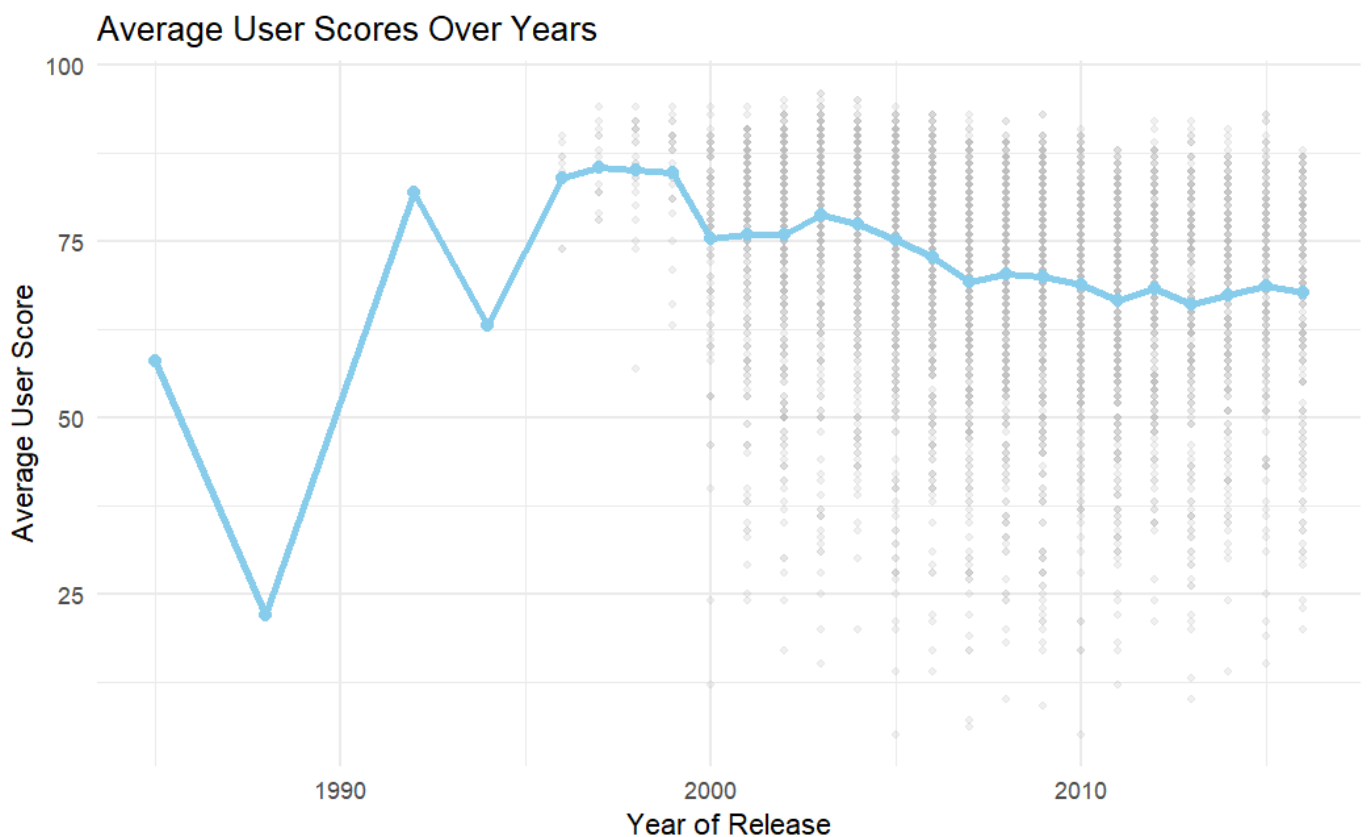
# Filter only User_Score
long_user <- long %>% filter(Type == "User_Score")
avg_year_user <- avg_year %>% filter(Type == "User_Score")

# Plot
score_over_year_plot <- ggplot(avg_year_user, aes(x = Year_of_Release, y = mean_score)) +
  # Background scatter
  geom_point(data = long_user, aes(x = Year_of_Release, y = Score),
            alpha = 0.2, size = 1, color = "grey") +
  # Average line
  geom_line(size = 1.2, color = "skyblue") +
  geom_point(size = 2, color = "skyblue") +
  labs(x = "Year of Release",
       y = "Average User Score",
       title = "Average User Scores Over Years") +
  theme_minimal()

ggsave("plots/score_over_year_plot.png", score_over_year_plot, width = 8.74, height = 5.11, d
pi = 300)

score_over_year_plot

```



# Model

$$Y = \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \beta_3 \cdot X_3 + \epsilon$$

where

$Y$  : User Score  
 $X_1$  : Critic Score  
 $X_2$  : Year of Release  
 $X_3$  : Global Sales

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```
model <- lm(User_Score ~ ., data = clean_data)

summary(model)
```

```
Call:
lm(formula = User_Score ~ ., data = clean_data)

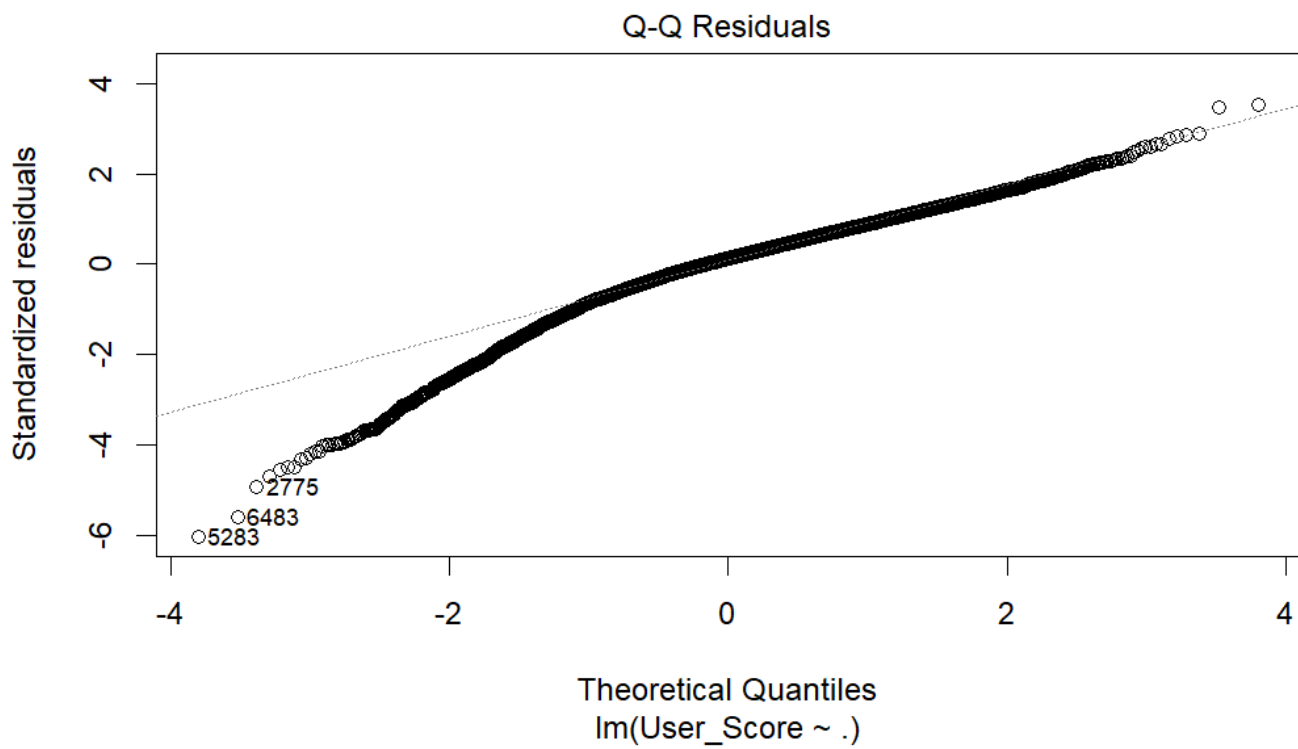
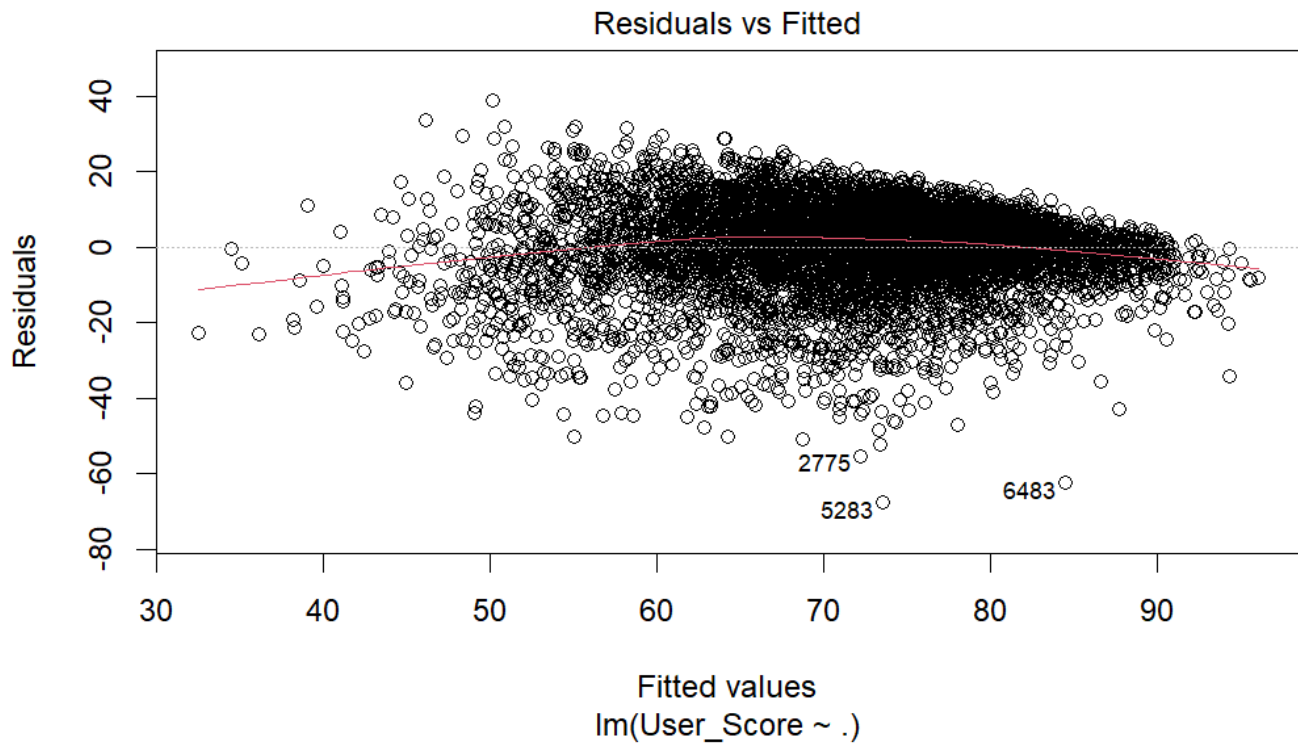
Residuals:
    Min       1Q   Median       3Q      Max
-67.561  -5.445   1.434   7.233  38.883

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.699e+03  6.377e+01  26.644 < 2e-16 ***
Critic_Score    6.117e-01  9.991e-03  61.227 < 2e-16 ***
Year_of_Release -8.318e-01  3.176e-02 -26.192 < 2e-16 ***
Global_Sales   -3.728e-01  7.084e-02  -5.263 1.46e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

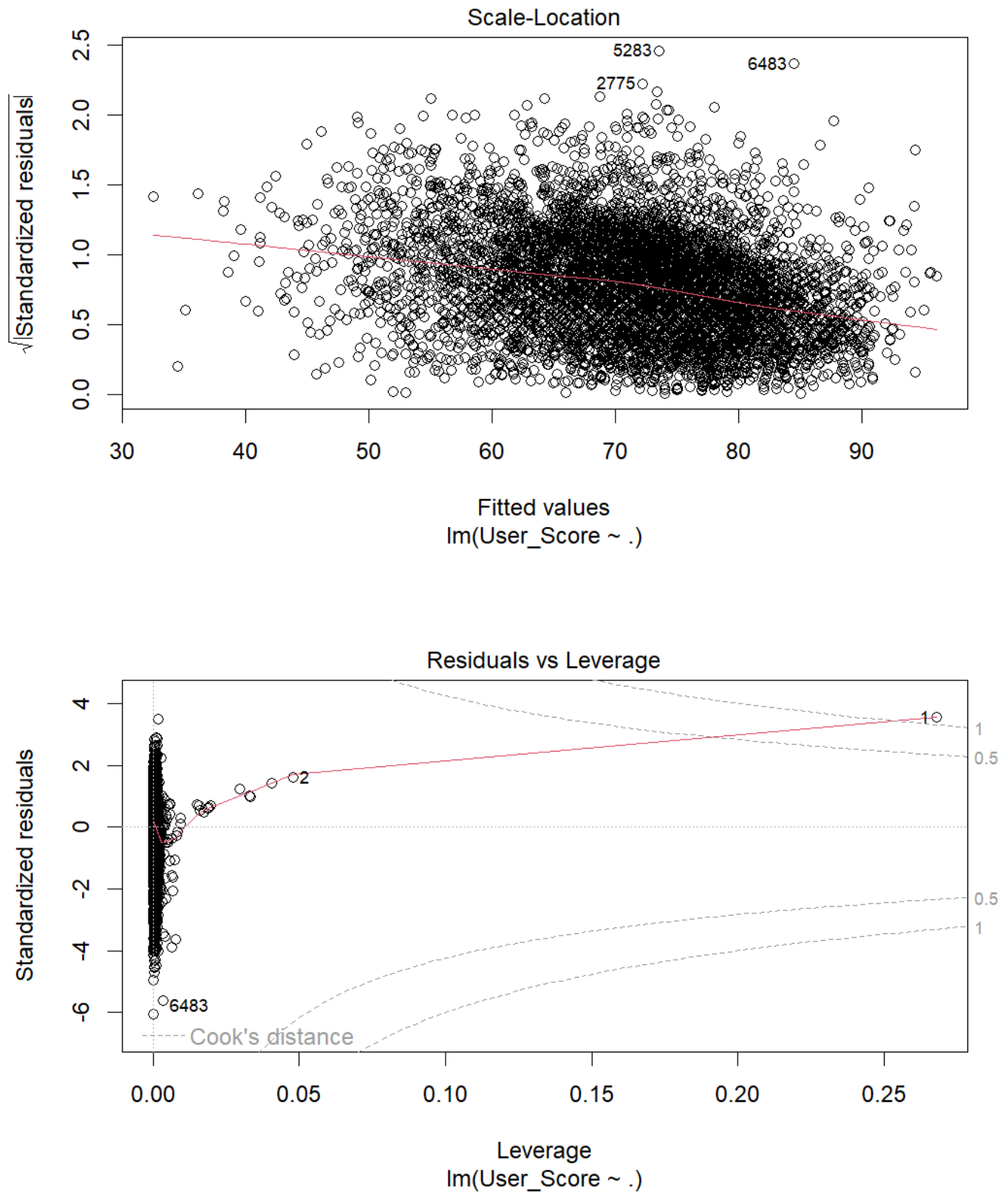
Residual standard error: 11.17 on 6890 degrees of freedom
Multiple R-squared:  0.3985,    Adjusted R-squared:  0.3982
F-statistic: 1521 on 3 and 6890 DF,  p-value: < 2.2e-16
```

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```
# Dont know how relevant this is
plot(model)
```







## ChatGPT Analysis

This is the summary of a multiple linear regression where you model `User_Score` using all other variables in `clean_data`:

`User_Score ~ Critic_Score + Year_of_Release + Global_Sales`

`User_Score~Critic_Score+Year_of_Release+Global_Sales`

I'll break it down piece by piece and then interpret it in plain language.

**1** Model fit (big picture) R-squared = 0.3985

About 40% of the variation in User\_Score is explained by the three predictors.

This is moderate explanatory power for real-world data.

Adjusted R-squared = 0.3982

Very close to  $R^2 \rightarrow$  adding predictors is justified.

F-statistic ( $p < 2.2e-16$ )

The model as a whole is highly statistically significant

At least one predictor meaningfully explains User\_Score

**2** Residuals (errors) Min -6.76 1Q -0.54 Median 0.14 3Q 0.72 Max 3.89

Errors are centered close to 0  $\rightarrow$  good

Slight right skew (larger positive max)

Typical prediction error  $\approx \pm 1.1$  User\_Score points

**3** Coefficients (core interpretation)  $\blacklozenge$  Intercept: 169.9 (Intercept) = 169.9

The predicted User\_Score when:

Critic\_Score = 0

Year\_of\_Release = 0

Global\_Sales = 0

Not meaningful in practice, but necessary for the model.

$\blacklozenge$  Critic\_Score: +0.061 Estimate = 0.061  $p < 2e-16$

 Interpretation:

For every 1-point increase in Critic\_Score, User\_Score increases by  $\sim 0.06$  points, holding other variables constant.

Very strong and significant predictor.

$\blacklozenge$  Year\_of\_Release: -0.083 Estimate = -0.083  $p < 2e-16$

 Interpretation:

Each additional year is associated with a 0.083 decrease in User\_Score.

Suggests older games have slightly higher user scores (possible nostalgia or selection bias).

Statistically very strong.

$\blacklozenge$  Global\_Sales: -0.037 Estimate = -0.037  $p = 1.46e-07$

 Interpretation:

For each additional million units sold, User\_Score drops by  $\sim 0.037$  points.

Small effect, but statistically significant.

Could reflect popular games appealing to broader (not more critical) audiences.

**4** Statistical significance

All predictors:

Have very low p-values

Are statistically significant at any conventional level

But ⚠:

Statistical significance  $\neq$  practical importance

Critic\_Score matters much more than Global\_Sales.

#### 5 What this model says in plain English

User ratings are strongly aligned with critic ratings. Older games tend to receive slightly higher user scores. Higher sales are weakly associated with lower user scores. The model explains about 40% of user score variability.

#### 6 Important caveats

Correlation  $\neq$  causation

Year\_of\_Release likely captures:

changes in rating culture

survivorship bias

Global\_Sales is often right-skewed  $\rightarrow$  log-transform may help

$\text{clean\_data} \log_{10} \text{sales} < -\log_{10} p(\text{clean\_data} \text{Global\_Sales})$